Advanced ops · my notes L13

when the built-ins are not enough

My own collector 🧰

Acc::new
empty: count 0
sum 0.00

→

Acc::add
count++, sum, min, max

→

Acc::merge
parallel only!
empty side? handle it

→

finish
→ record
MoneyStats

→ count=12 sum=4429.84 min=18.80 max=1500.00 avg=369.15 · parallel = same ✓

Streams from nowhere 🌱

`iterate(Aug, ≤ Nov, +1) [9]`
2026-08 … 2026-11

`generate(..).limit(3)`
------ ------ ------

`concat(DEP, ARR)`
DEP SQ318 … ARR GA827

deep: customers ⟶ accounts ⟶ transactions (flatMap × 2) → 27 for Singapore

no limit, no predicate → never ends!

merge / compute 🗂

`totals.merge(id, bal, add)`

`computeIfAbsent(id, k -> new ArrayList<>())`

one by one → in place
whole list → stream toMap

Map → new map 🔁

`entrySet().stream()`

sort by value → **LinkedHashMap**
top 3: C003, C010, C006
HashMap loses the order

Lazy paging 📄

Iterator → Spliterator →
`StreamSupport.stream`
count() → fetches=4
findFirst() → fetches=1

Beyond Java 21 🔭

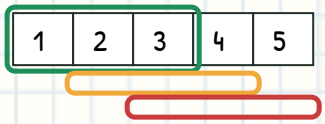
Gatherers (Java 24): windowFixed · windowSliding · scan
this course: IntStream indexes + loops

Batches, windows, running totals

chunk(1..250, 100):



windows([1..5], 3):



[[1,2,3], [2,3,4], [3,4,5]]

running total: 3314.36 → 9412.30 → 15244.95 loop ✓ O(n) acc[0] in map ×

Worked example: payment statistics

BEFORE: 4 hand-synced HashMaps · AFTER [8]: 4 groupingBy passes + 2 Optionals · MODERN: `groupingBy(method, TreeMap::new, mapping(amount, moneyStats()))` · CARD: count=5 sum=2070.29 … avg=414.06 · all 3 agree ✓

Exercise A: running balance 🏦

opening = closing − net SGD movements · ACC-1014: OPENING 19235.05 → T001 −150.39 = 19084.66 → … → CLOSING 18900.00 · lowest ACC-1013: −4382.09

Exercise B: payment-file batches 📦

batches of 5: P001..P005 2065.65 · P006..P010 1954.19 · P011..P012 410.00 · control totals = 4429.84 ✓ · batches of 100: 1 batch of 12

★ Remember

Collector.of + test parallel · limit infinite streams · merge in place, streams for new maps · indexes for batches · loops for running totals · lazy paging · next → L14 parallel streams